ECM3412

UNIVERSITY OF EXETER

SCHOOL OF ENGINEERING, COMPUTER SCIENCE AND MATHEMATICS

Nature-Inspired Computation

TWO HOURS

Answer question 1, and two out of the four other questions.
Question 1 is worth 40 marks. Other questions are worth 30 marks each.

Candidates are advised to spend FORTY-FIVE minutes on question 1 and THIRTY-FIVE minutes on each of the other questions.

No electronic calculators of any sort are to be used during the course of this examination.

Compulsory question

1. (a) Write pseudocode for *particle swarm optimisation* (PSO). Explain how each variable used affects the ability of PSO to find a near-optimal solution.

(8 marks)

(b) Describe what is meant by an *approximate* algorithm and state why this type of algorithm is required to optimise most real-world problems.

(6 marks)

(c) In evolutionary algorithms, selection is an important part of the algorithm procedure.

i. Describe the process of *ranked roulette wheel* selection.

(4 marks)

ii. Explain how the selection pressure can be modified in the ranked roulette wheel selector.

(2 marks)

(d) Multi-objective optimisation makes use of the principle of *domination*.

i. Give a definition of dominance in multi-objective optimisation

(3 marks)

ii. Explain what is meant by the statement ‘Solution A is on the non-dominated (Pareto) front’

(3 marks)

(e) Explain how *virtual pheremone* aids in the search for the shortest path in ant colony optimisation.

(6 marks)

(f) Explain the main differences between a *multilayer perceptron* and a *self-organising map*. In your answer make sure you refer to the architecture, the learning algorithm and the problem types that each can be used for.

(8 marks)

(Total 40 marks)

Please Turn Over

2. (a) Cellular automata are widely used to simulate natural phenomena. Explain the following cellular automata terms and their relationship to one another (with the aid of diagrams where necessary):

i. Cell State
ii. Cell Neighbourhood
iii. State Transition Rules
iv. Discrete Time
v. Toroid

(15 marks)

(b) Genetic programming can be used in conjunction with cellular automata. Describe how you would combine the two algorithms to optimise a simulation of your choosing. In particular, provide detail on the following aspects:

i. The representation of solutions
ii. The creation of random individuals
iii. The fitness function used to evaluate individuals

(15 marks)

(Total 30 marks)

Please Turn Over

3. The results obtained from evolutionary algorithms are sensitive to a number of algorithm parameters. Explain how the following parameter settings will affect the ability of the algorithm to find a good solution, and its speed of execution.

(a) The population size

(4 marks)

(b) The number of iterations

(4 marks)

(c) The mutation rate

(4 marks)

(d) The selection pressure

(4 marks)

A further important aspect in the use of evolutionary algorithms is the *representation*.

(e) Explain what is meant by the terms *direct* and *indirect* representations. Give an example of each.

(6 marks)

(f) The representation can affect the choice of mutation and crossover operators. Explain why this is the case and describe the operators necessary to optimise the travelling salesman problem.

(8 marks)

(Total 30 marks)

Please Turn Over

4. Ant colony optimisation can be used for a number of different applications.

(a) Describe the main algorithm and, briefly, the experimental findings in the paper Di Caro G., Dorigo M. (1998) ”Ant Colonies for Adaptive Routing in Packet-Switched Communications Networks” in the *Proceedings of PPSN - Lecture Notes in Computer Science* Vol. 1498, pp. 673-682.

(15 marks)

(b) Describe the process of using ant colony optimisation to find a solution to a University timetabling problem, where the University timetabling problem is defined as follows:

The timetabling of E exams sat by S students in P periods during the week, subject to the requirements that: no student should be timetabled for more than one exam at one time; where possible, students should not be scheduled for exams in consecutive periods; and final period exams should be avoided.

In your answer, make sure that you discuss the representation of the problem and give a detailed account of how the algorithm operates, along with a description of the experiments you would conduct to assess its performance.

(15 marks)

(Total 30 marks)

Please Turn Over

5. Neural networks can be used for a variety of data processing and classification tasks.

(a) Explain what is meant by an *activation function* in a neural network.

(3 marks)

(b) Give an example of a commonly used activation function and, with the aid of a diagram explain how the function works.

(3 marks)

(c) Explain what is meant by a *weight* and how it is crucial to the operation of neural networks.

(3 marks)

(d) Explain, with the aid of equations, how a multilayer perceptron:

i. Calculates an output given a set of input data.

(7 marks)

ii. Calculates the error on the output and updates the weights to generate a closer output in the next iteration.

(8 marks)

(e) During learning, neural networks can suffer from *overfitting*. Explain what this term means and what steps can be taken to reduce its effect or stop it from occurring altogether.

(6 marks)

(Total 30 marks)

End of Paper